



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Name of Student: Sumit Kr. Mahato

PRN: 25070122195

Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 4

TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

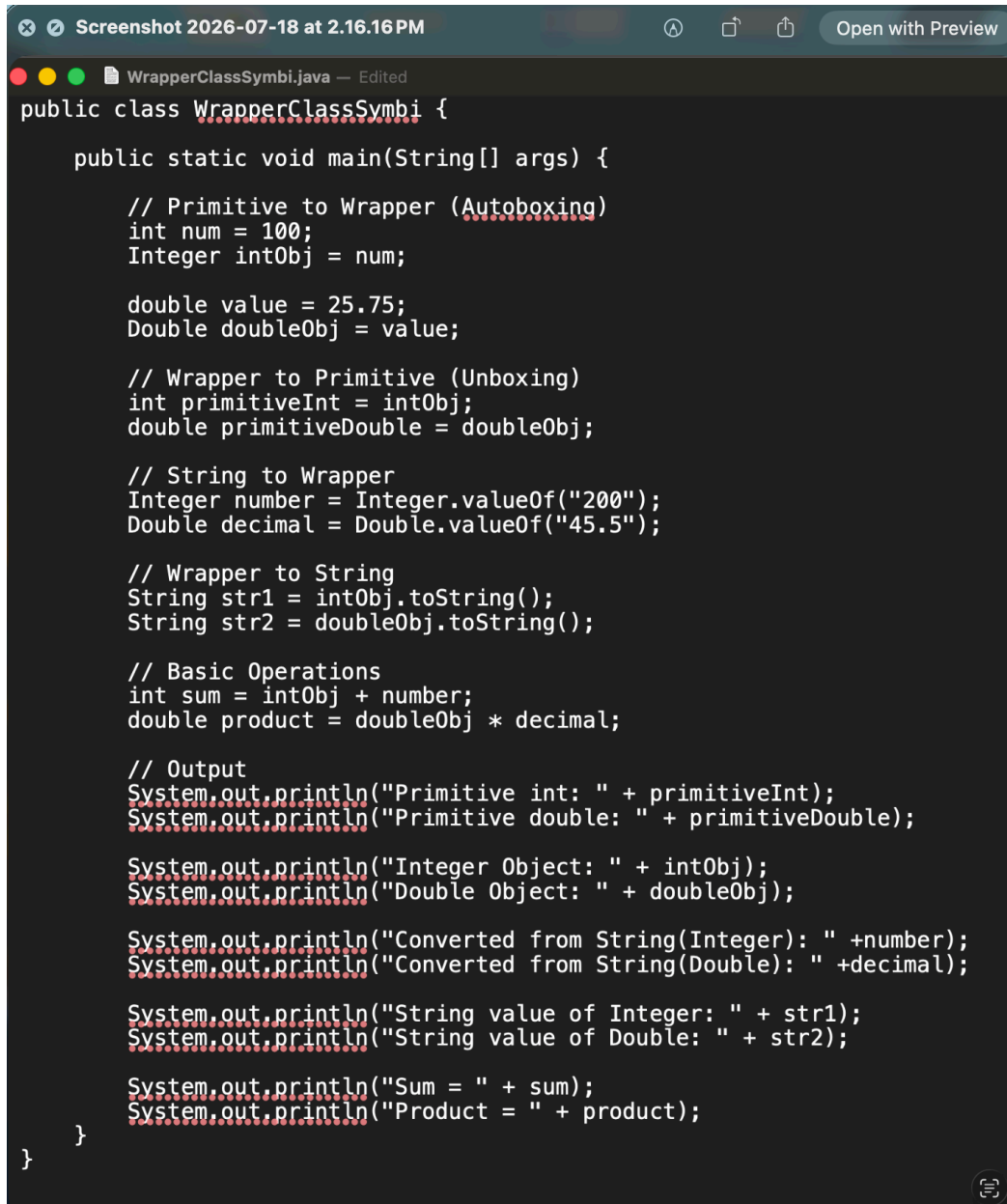
Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Example Code:



```
Screenshot 2026-07-18 at 2.16.16 PM
WrapperClassSymbi.java — Edited

public class WrapperClassSymbi {

    public static void main(String[] args) {

        // Primitive to Wrapper (Autoboxing)
        int num = 100;
        Integer intObj = num;

        double value = 25.75;
        Double doubleObj = value;

        // Wrapper to Primitive (Unboxing)
        int primitiveInt = intObj;
        double primitiveDouble = doubleObj;

        // String to Wrapper
        Integer number = Integer.valueOf("200");
        Double decimal = Double.valueOf("45.5");

        // Wrapper to String
        String str1 = intObj.toString();
        String str2 = doubleObj.toString();

        // Basic Operations
        int sum = intObj + number;
        double product = doubleObj * decimal;

        // Output
        System.out.println("Primitive int: " + primitiveInt);
        System.out.println("Primitive double: " + primitiveDouble);

        System.out.println("Integer Object: " + intObj);
        System.out.println("Double Object: " + doubleObj);

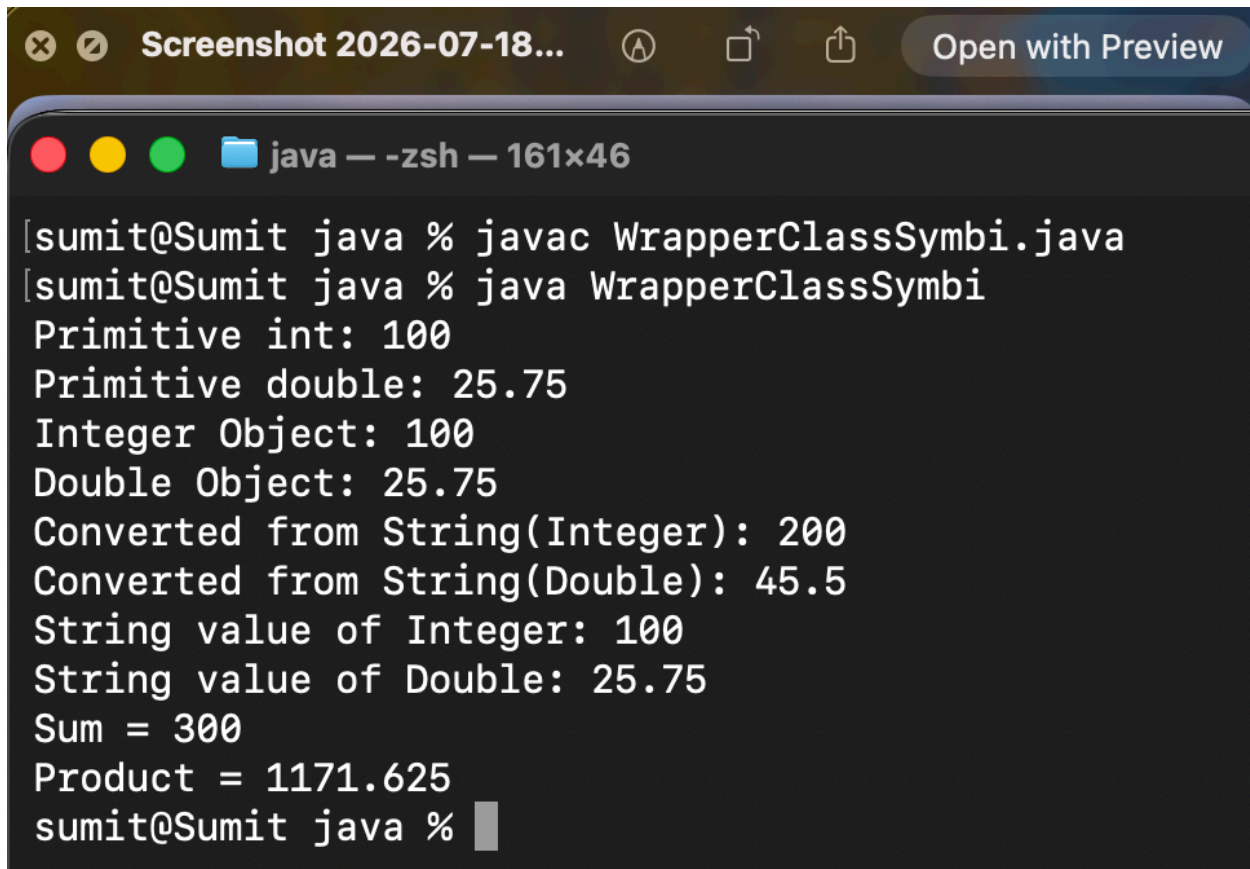
        System.out.println("Converted from String(Integer): " + number);
        System.out.println("Converted from String(Double): " + decimal);

        System.out.println("String value of Integer: " + str1);
        System.out.println("String value of Double: " + str2);

        System.out.println("Sum = " + sum);
        System.out.println("Product = " + product);

    }
}
```

Output:



The image shows a screenshot of a macOS terminal window. The title bar at the top reads "Screenshot 2026-07-18..." and includes standard window controls (close, zoom, full screen) and a button that says "Open with Preview". The terminal window itself has a title bar that says "java — -zsh — 161x46". The terminal content shows the following sequence of commands and output:

```
[sumit@Sumit java % javac WrapperClassSymbi.java  
[sumit@Sumit java % java WrapperClassSymbi  
Primitive int: 100  
Primitive double: 25.75  
Integer Object: 100  
Double Object: 25.75  
Converted from String(Integer): 200  
Converted from String(Double): 45.5  
String value of Integer: 100  
String value of Double: 25.75  
Sum = 300  
Product = 1171.625  
sumit@Sumit java %
```

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

Source Code:

```
Users > sumit > Documents > java > java1 > StudentMarks.java > StudentMarks
1  class StudentMarks
2  {
3      Run | Debug
4      public static void main(String[] args)
5      {
6          String m1 = "85";
7          String m2 = "90";
8          String m3 = "78";
9
10         Integer mark1 = Integer.valueOf(m1);
11         Integer mark2 = Integer.valueOf(m2);
12         Integer mark3 = Integer.valueOf(m3);
13
14         int total = mark1 + mark2 + mark3;
15
16         System.out.println("Mark 1: " + mark1);
17         System.out.println("Mark 2: " + mark2);
18         System.out.println("Mark 3: " + mark3);
19         System.out.println("Total Marks: " + total);
20     }
```

Output:

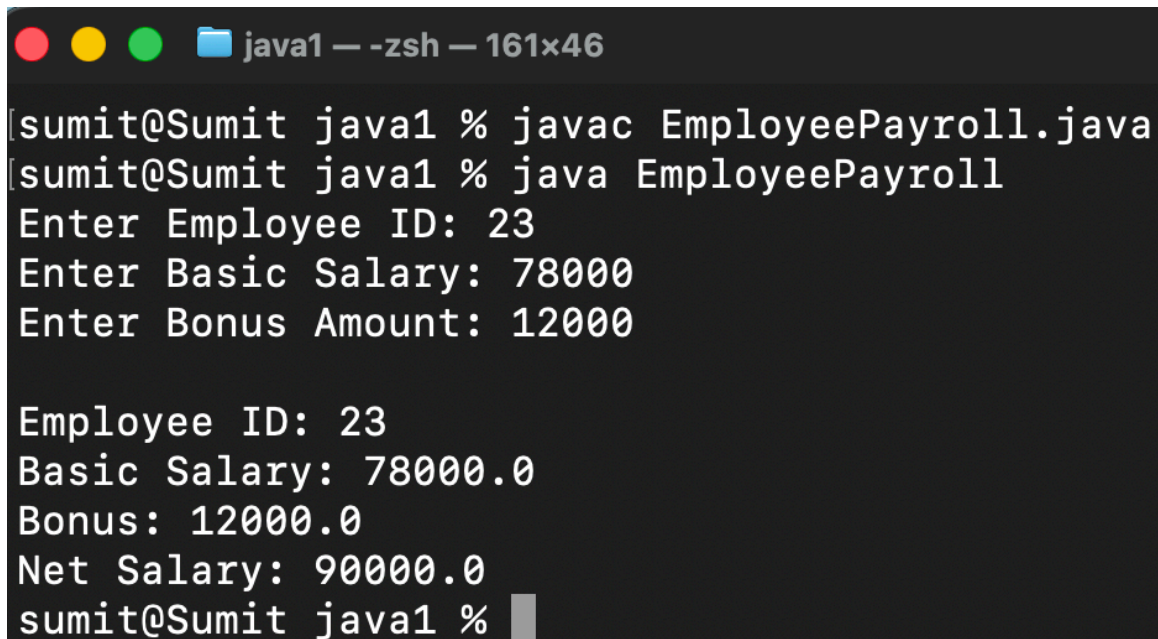
```
java1 — -zsh — 161x46
[sumit@Sumit java1 % javac StudentMarks.java
[sumit@Sumit java1 % java StudentMarks
Mark 1: 85
Mark 2: 90
Mark 3: 78
Total Marks: 253
sumit@Sumit java1 %
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

Source Code:

```
Users > sumit > Documents > java > EmployeePayroll.java > EmployeePayroll > main(String[])
1  import java.util.Scanner;
2  class EmployeePayroll
3  {
    Run | Debug
4      public static void main(String[] args)
5      {
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print("Enter Employee ID: ");
9          String id = sc.nextLine();
10
11         System.out.print("Enter Basic Salary: ");
12         String salary = sc.nextLine();
13
14         System.out.print("Enter Bonus Amount: ");
15         String bonus = sc.nextLine();
16
17         Integer empId = Integer.valueOf(id);
18         Double basicSalary = Double.valueOf(salary);
19         Double bonusAmount = Double.valueOf(bonus);
20
21         if (basicSalary <= 0)
22         {
23             System.out.println("Invalid Salary!");
24         }
25         else
26         {
27             double netSalary = basicSalary + bonusAmount;
28
29             System.out.println("\nEmployee ID: " + empId);
30             System.out.println("Basic Salary: " + basicSalary);
31             System.out.println("Bonus: " + bonusAmount);
32             System.out.println("Net Salary: " + netSalary);
33         }
34         sc.close();
35     }
36 }
```

Output:



```
java1 — -zsh — 161x46

[sumit@Sumit java1 % javac EmployeePayroll.java
[sumit@Sumit java1 % java EmployeePayroll
Enter Employee ID: 23
Enter Basic Salary: 78000
Enter Bonus Amount: 12000

Employee ID: 23
Basic Salary: 78000.0
Bonus: 12000.0
Net Salary: 90000.0
sumit@Sumit java1 %
```

Conclusion:

This assignment demonstrated the use of Java Wrapper Classes for converting between String, primitive data types, and wrapper objects. It showed how methods like `valueOf()` and autounboxing are used to perform conversions and basic operations such as calculating total marks and net salary. It also highlighted the importance of wrapper classes in handling object-based data and validating input values in Java applications.